

Vision-Language Models for Embodied Reasoning

Abstract

The integration of multimodal foundation models into physical agents has catalyzed a paradigm shift in artificial general intelligence, yielding a novel class of systems known as Vision-Language-Action (VLA) architectures. This comprehensive survey provides an exhaustive examination of VLA models engineered for embodied decision-making, exploring how these systems transition from passive observers of internet-scale data to active, physically grounded participants in complex environments. Historically, robotic control relied on rigid, modular pipelines segregating perception, task planning, and motor execution. Contemporary VLA systems upend this paradigm by unifying these modalities, directly mapping natural language instructions and raw visual observations into low-level kinematic commands. This manuscript systematically categorizes the rapidly evolving landscape of embodied foundation models. It explores monolithic end-to-end architectures that leverage discrete tokenization or continuous flow-matching, dual-system hierarchical designs that decouple high-level logic from low-level control, and the critical implementation of 3D spatial comprehension mechanisms. Furthermore, the survey investigates predictive world models that simulate future environmental states and active perception frameworks that allow robots to dynamically optimize their sensor viewpoints. Core application domains, including complex robotic manipulation, continuous Vision-Language Navigation (VLN), and Embodied Question Answering (EQA), are analyzed in depth. Finally, the analysis details pressing deployment challenges, spanning the sim-to-real data gap, hardware-constrained edge execution, catastrophic representation degradation during fine-tuning, and the security vulnerabilities inherent in generative physical simulators. By synthesizing findings from leading architectures, this work outlines the structural roadmap for the future of generalist robotic policies.

Introduction

The theoretical foundation of embodied intelligence posits that true artificial general intelligence (AGI) cannot be achieved solely through abstract symbolic computation; it inherently requires an agent to perceive, interpret, and dynamically interact with the physical world¹. For decades, the field of robotics operated under paradigms dominated by classical Task and Motion Planning (TAMP) and deep reinforcement learning (RL)². While classical approaches excelled in highly structured, deterministic environments such as industrial manufacturing floors, they consistently failed when subjected to the open-ended variability, partial observability, and semantic ambiguity of human-centric spaces. Conversely, deep RL approaches demonstrated remarkable adaptability but suffered from extreme sample inefficiency and a profound lack of generalizability across different tasks and robotic embodiments².

The rapid maturation of Large Language Models (LLMs) and Vision-Language Models (VLMs)

introduced unprecedented semantic comprehension, zero-shot generalization, and open-vocabulary classification to computational systems⁵. Models pre-trained on internet-scale corpora demonstrated emergent reasoning capabilities, including chain-of-thought deduction and the ability to parse complex human instructions⁷. However, adapting these conversational and passive-perception models to physical hardware requires sensorimotor grounding—the ability to seamlessly map abstract linguistic concepts and two-dimensional pixel arrays to continuous, high-frequency physical action spaces⁸. This operational necessity has driven the creation of Vision-Language-Action (VLA) models, a novel class of multimodal foundation models designed to natively process information from vision, language, and action². Pioneered by architectures such as Google DeepMind's Robotic Transformer 2 (RT-2), VLAs bypass traditional modular pipelines¹². Rather than utilizing isolated, disjointed algorithms for object detection, trajectory planning, and inverse kinematics, end-to-end VLAs ingest massive datasets of image-text pairs alongside millions of robotic demonstration trajectories. This unified training enables robust, multi-task generalization across disparate robot configurations, allowing agents to manipulate unseen objects based on abstract semantic instructions¹³.

Despite these remarkable advances, scaling physical AI presents unique architectural, geometric, and systemic hurdles. The physical world is inherently three-dimensional, highly dynamic, and computationally demanding. Agents must maintain object permanence, predict continuous physical dynamics, and actively maneuver their sensors to resolve environmental occlusions¹⁶. Consequently, the research landscape is rapidly expanding beyond simple monolithic action-prediction networks. Researchers are increasingly integrating predictive world models that hallucinate future states to guide immediate actions¹⁹, implementing 3D spatial encoding to improve geometric comprehension²², and deploying hierarchical structures that effectively isolate high-level strategic planning from low-level, high-frequency motor control²⁴.

This survey provides a detailed taxonomy of the architectures, training paradigms, and applications driving modern embodied AI². Through an exhaustive comparative analysis of the current literature, this manuscript outlines the underlying mechanics of physical foundation models, evaluates their performance across standardized benchmarks, and identifies the critical technological barriers that must be overcome to achieve robust, universally deployable autonomous systems.

Methods

The methodologies underlying embodied foundation models can be categorized into distinct architectural paradigms, ranging from heavily integrated monolithic networks to decoupled, hierarchical systems supported by internal environmental simulators.

Vision-Language-Action (VLA) Monolithic Architectures

Monolithic VLA models represent the dominant, end-to-end approach to general-purpose robot control. These systems treat vision, language, and action as a unified sequence of tokens processed by a single transformer-based network²⁶. By leveraging the immense pre-trained

weights of underlying VLMs, these architectures transfer internet-scale semantic knowledge directly into motor control. This allows robots to interpret novel instructions and manipulate unseen objects without task-specific programmatic rules²⁵. The defining characteristic differentiating leading VLA models is their methodology for action representation and kinematic generation. Actions in the physical world are strictly continuous, whereas standard language models are optimized to output discrete tokens over a fixed vocabulary. Early prominent VLAs, such as RT-2 and OpenVLA, rely on autoregressive discrete tokenization¹². In this formulation, continuous action spaces are quantized into discrete bins, often allocating 256 bins per physical dimension. The model then predicts action tokens sequentially, operating analogously to a language model generating text. OpenVLA, a 7-billion-parameter open-source model, utilizes a Prismatic VLM backbone combining Llama 2 with dual visual encoders: SigLIP for extracting dense semantic features and DINOv2 for capturing spatial geometry²⁹. Trained on 970,000 real-world trajectories from the Open X-Embodiment dataset, OpenVLA demonstrates extreme robustness, significantly outperforming closed-source counterparts like the 55-billion-parameter RT-2-X across diverse tasks²⁹. However, while discrete autoregressive prediction is architecturally simple, it frequently results in jerky, single-step kinematic movements that lack the fluidity required for complex, high-frequency manipulation¹².

To address the latency and kinematic instability of single-step autoregression, models such as SmolVLA predict a sequence of future actions simultaneously, a technique known as chunked action prediction. By forecasting 10 to 50 steps into the future, the model ensures smoother trajectories, reduces the autoregressive inference bottleneck, and maximizes compute efficiency, allowing for deployment on consumer-grade hardware³⁰.

Moving entirely away from discrete tokens, advanced architectures utilize diffusion-based action decoders and flow-matching formulations¹². The Octo model, a lightweight open-source generalist policy, employs a diffusion policy that predicts a distribution of possible continuous actions, iteratively denoising a random sample to uncover the optimal trajectory¹². This is highly advantageous in multimodal environments where multiple valid paths exist to achieve identical

goals. Further advancing this, the π_0 (pi-zero) model employs a sophisticated flow-matching

VLA architecture¹². Fusing a pre-trained PaliGemma VLM with an action expert, π_0 generates high-frequency continuous actions at up to 50 Hz. By utilizing flow-matching to iteratively denoise samples into valid trajectories, it produces exceptionally smooth physical movements capable of supporting complex bimanual coordination across disparate robotic arms¹². A

recent extension, π_0 -FAST, further optimizes this by leveraging Frequency-space Action Sequence Tokenization, converting continuous tokens into the frequency domain via discrete cosine transforms to achieve unprecedented compression¹².

Model Framework	Backbone / Architecture	Vision Encoder	Action Representatio	Key Distinguishing
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			n Strategy	Feature
RT-2 [cite: 30]	PaLI-X / PaLM-E	ViT-e	Discrete Tokens (Autoregressive)	First large-scale validation of emergent semantic generalization in physical robotic tasks.
OpenVLA [cite: 29]	Llama 2 (7B)	SigLIP + DINOv2	Discrete Tokens (Autoregressive)	Open-source state-of-the-art; dual vision encoders for combined spatial and semantic grounding.
UniVLA [cite: 28]	Unified Sequence Transformer	Shared Vocabulary	Discrete Tokens (Autoregressive)	Unifies vision, text, and action into a single shared discrete vocabulary for massive video pre-training.
Octo [cite: 12]	Lightweight Transformer	CNN-based	Continuous (Diffusion Policy)	Parameter-efficient open-source generalist policy; highly adaptable to novel sensor setups.
π_0 [cite: 12]	PaliGemma	SigLIP + Gemma	Continuous (Flow-Matching)	State-of-the-art high-frequency

			g)	y continuous control; superior handling of multimodal action distributions.
RoboMamba [cite: 32]	Mamba SSM (Linear)	CLIP	Continuous (Policy Head)	Utilizes State Space Models for extremely fast inference without the quadratic attention costs of transformers.

Hierarchical and Dual-System Planning Frameworks

While monolithic VLAs excel in short-horizon atomic tasks, their reliance on massive paired instruction-vision-action datasets severely limits their out-of-distribution capabilities and logical endurance²⁵. End-to-end models frequently collapse when tasked with long-horizon objectives requiring sequential logical deduction, as any early mistake compounds uncontrollably over time²⁴. To circumvent this strict data dependency, researchers have developed hierarchical, or dual-system, VLA architectures³³.

These systems explicitly decouple cognitive planning from sensorimotor execution²⁴. A high-level semantic planner (analogous to System 2 cognitive processing) evaluates the overarching goal and current visual context, decomposing a complex instruction into a sequence of executable sub-goals³⁴. A separate, low-level control policy (System 1) then executes these atomic sub-goals⁴. Frameworks like SayCan, Inner Monologue, and ProgPrompt pioneered this approach by utilizing LLMs to formulate executable robot policies that respect environmental constraints³⁵.

Advanced interactive planners, such as the DEPS (Describe, Explain, Plan, and Select) framework, integrate real-time feedback loops. DEPS requires the LLM to generate an initial plan, monitor physical execution, and provide self-generated explanations if an action fails³⁶. This self-reflection allows the system to dynamically select alternative strategies upon encountering failures, effectively doubling the success rate on open-ended tasks compared to static planners³⁶. Similarly, architectures like CoPAL and RL-GPT leverage LLMs for top-down strategy formulation while delegating fine-grained control to task-specific Reinforcement

Learning (RL) policies, vastly improving sample efficiency³⁷.

The primary architectural friction in hierarchical design lies in formulating the intermediate representation bridging the two systems. Passing dense visual representations, such as goal images, provides rich information but traditionally requires the low-level policy to be explicitly trained on paired action data to interpret them, nullifying the zero-shot objective²⁵. Conversely, sparse symbolic representations often lack the strict geometric detail required for precise manipulation²⁵. To resolve this, recent hierarchical models utilize object-centric representations or visual Chain-of-Thought (CoT) mechanisms to align high-level strategy with precise spatial constraints, ensuring the planner does not issue commands that violate physical geometry²⁵.

3D Spatial Comprehension and Representations

Physical reality is inherently three-dimensional, yet the vast majority of VLMs are pre-trained exclusively on 2D internet images. When tasked with embodied execution, standard 2D MLLMs exhibit profound deficits in spatial reasoning, lacking the ability to maintain object permanence, infer depth, or track coordinates consistently across multiple egocentric viewpoints¹⁶. An embodied agent must understand depth, occlusion, and layout to safely navigate and manipulate its surroundings.

To address this critical gap, researchers are injecting 3D spatial data directly into language models, establishing the "3D-LLM" paradigm⁶. One foundational approach involves Language Embedded Radiance Fields (LERFs), which distill dense semantic embeddings from models like CLIP directly into Neural Radiance Fields (NeRFs) during volume rendering⁴¹. This fusion permits the querying of volumetric 3D spaces using open-ended natural language, allowing a robot to locate targets with high spatial precision irrespective of the camera angle⁴². Similarly, ConceptFusion enables zero-shot spatial reasoning by aligning language models with 3D scene graphs, outperforming supervised baseline models in 3D Intersection-over-Union metrics by substantial margins⁴².

Other methodologies project 3D point clouds directly into language embedding spaces. Scene-LLM, for instance, adopts a unified 3D visual feature representation that incorporates both egocentric viewpoints and global scene contexts⁴². This dual-perspective approach is vital for localizing the agent while simultaneously calculating holistic environmental changes during interactive planning tasks⁴³. The MLLM-For3D architecture leverages powerful 2D MLLMs to generate multi-view pseudo-segmentation masks. It applies token attention mechanisms to filter out irrelevant frames and employs a token-for-query approach to unproject 2D masks into a coherent 3D semantic space, strictly enforcing geometric consistency⁴⁰.

In the realm of direct motor control, the SpatialVLA model integrates an "Ego3D Position Encoding" module to fuse 3D spatial context directly with semantic features extracted by a SigLIP encoder²². SpatialVLA further discretizes continuous movement into "Adaptive Action Grids" derived from statistical distributions of real-world trajectories. This native 3D grounding significantly enhances the model's accuracy in tasks requiring precise spatial configurations, proving highly superior to flat 2D representations when tested on complex out-of-distribution positional changes²².

Predictive Dynamics and World Models

A critical divergence between simple reactive algorithms and advanced biological intelligence is the capacity to simulate hypothetical futures before committing to an action. In embodied AI, this capability is achieved through "World Models"—learned internal simulators that capture the structural and dynamic transition probabilities of a physical environment¹³. World models enable agents to perform sample-efficient planning, counterfactual deduction, and multi-step reasoning without requiring expensive or dangerous physical trial-and-error¹⁹.

The evolution of world models has transitioned from purely latent dynamics (e.g., Dreamer, MuZero) to massive, generative video models (e.g., GAIA-1, UniSim, Genie)¹⁹. These foundation-scale models ingest current visual states alongside proposed action tokens to render highly realistic video predictions of the subsequent state⁹. However, unconstrained video generation alone is often insufficient for rigorous robotic control, as pixel-level predictions can blur or hallucinate impossible physics over extended temporal horizons³. To enforce stricter structural logic, the FactorSim framework generates fully functional, mathematically rigorous Partially Observable Markov Decision Process (POMDP) simulations from natural language prompts⁴⁷. By utilizing a model-view-controller software paradigm and systematically restricting the prompt context at each generation step via a factor graph, FactorSim produces interpretable, coded simulation environments suitable for training reliable RL policies without hallucinating non-existent functions⁴⁷.

In direct VLA integration, the ForeAct (Visual Foresight Planning) framework utilizes a deeply compressed autoencoder and a linear Diffusion Transformer (DiT) derived from the SANA

architecture to generate high-resolution (640×480) future observations⁵¹. Running in merely 0.33 seconds on modern accelerator hardware, ForeAct feeds this "imagined" future state into the VLA. By providing a clear visual target of the completed subtask, the VLA is relieved of high-level semantic deduction and can devote all its parameters to precise visuo-motor execution, yielding a 40.9% absolute improvement over baseline flow-matching models in complex real-world benchmarks⁵².

Similarly, the Hierarchical World Model (H-WM) addresses the compounding degradation seen in long-horizon predictions by jointly predicting logical symbolic state transitions alongside visual latent features²¹. The high-level logical world model ensures strict adherence to physical constraints and globally consistent task ordering, while the visual world model grounds these abstract symbols into perceptual space²¹. This bilevel simulation tightly anchors abstract strategy to realistic environmental constraints, preventing the system from pursuing impossible trajectories³.

Active Perception and Directed Exploration

Traditional VLA frameworks largely rely on passive perception—executing decisions based on a static, pre-captured image⁵⁶. However, real-world environments are characterized by heavy occlusion, limited fields of view, and dynamic obstacles¹⁷. Active perception is the deliberate, autonomous adjustment of an agent's sensors to acquire task-critical information that is not

immediately visible⁵⁶.

The I-Perceive foundation model achieves active perception by fusing a VLM backbone with a geometric Visual General Geometry Transformer (VGGT)¹⁷. This integration bridges semantic intent with 3D structural comprehension, allowing a mobile manipulator to deduce precisely where to move its camera to satisfy open-ended linguistic queries¹⁷.

In fine-grained manipulation, the ActiveVLA framework utilizes a coarse-to-fine operational pipeline⁵⁸. Initially, it projects a 3D point cloud onto multiple 2D planes to identify critical regions. It then calculates an active viewpoint selection score, balancing geometric diversity and visibility by running KDTree-based ray-casting to ensure the proposed line of sight is unobstructed by scene geometry. Finally, the system executes an active 3D zoom-in on the target, providing high-resolution local features to the action decoder⁵⁸.

Similarly, the AP-VLM architecture projects a 3D virtual grid anchored to the robot's base frame over the scene. When queried, a VLM Perception Analyzer evaluates if the current view is sufficient. If insufficient, an Active Perception Policy queries the grid coordinates to calculate optimal rotational and translational shifts for the in-hand camera. This spatial anchoring allows the robot to autonomously peer inside overturned containers or look behind obstructions before initiating a physical grasp, effectively bridging high-level semantic reasoning with low-level spatial control⁵⁶.

Applications and Evaluation Benchmarks

The convergence of multimodal foundation models with robotic hardware has accelerated capabilities across several distinct application domains, each presenting unique evaluation metrics and environmental demands.

Complex Robotic Manipulation

Robotic manipulation remains the primary testbed for VLA architectures, demanding sub-millimeter precision, contact physics estimation, and complex spatial coordination³³. The creation of massive collaborative datasets, most notably the Open X-Embodiment dataset (encompassing over one million episodes across 22 distinct robot platforms), has been instrumental in training generalist policies capable of positive cross-embodiment transfer¹². Models are routinely evaluated on standardized suites such as CALVIN, LIBERO, and SimplerEnv, which test multi-task generalization and spatial reasoning²⁸. Research has rapidly expanded into bimanual coordination, where dual arms must move in continuous concert to fold garments or assemble components. In these scenarios, the fluid, high-frequency outputs of flow-matching action heads have proven essential for maintaining synchronization¹².

Vision-Language Navigation (VLN)

Vision-Language Navigation (VLN) requires an embodied agent to traverse complex, continuous environments by interpreting sequential natural language instructions and real-time visual streams⁶¹. A profound bottleneck in VLN is the quadratic computational cost of transformer attention mechanisms when processing long-horizon historical observations⁶³. To maintain temporal context without exhausting token limits, the Efficient-VLN architecture introduces dynamic memory management. It utilizes a progressive memory allocation system

that assigns higher token densities to recent frames while compressing distant historical context into a learnable recursive memory state⁶³.

Alternatively, systems like NavGemini manage token quotas by utilizing a hierarchical prompt manager, breaking the navigation task into localized sub-queries that consult the multi-modal LLM only when critical junction decisions are required⁶⁴. The domain has also expanded into Aerial VLN, focusing on Unmanned Aerial Vehicles (UAVs)⁶². Aerial VLN introduces severe complexities: continuous 6-DoF volumetric action spaces, extreme viewpoint variability due to altitude shifts, and strict power and weight limitations. Architectures such as the OpenFly framework address these challenges by creating keyframe-aware agents capable of grounding long instructions into city-scale trajectories while operating entirely within edge-hardware constraints⁶².

Embodied Question Answering (EQA)

Embodied Question Answering (EQA) merges spatial navigation with semantic comprehension. An agent must proactively explore an environment to gather the visual evidence necessary to answer a user's query⁶⁶. Standard VLMs generally lack the capacity for explicit intermediate planning, often resulting in excessive, undirected wandering. The FAST-EQA framework rectifies this by maintaining a bounded scene memory—a fixed-capacity set of region-target hypotheses that are updated online⁶⁶. This prevents unbounded memory growth while utilizing step-wise deductive logic to confine the physical search to question-relevant sub-spaces⁶⁶. A critical, yet historically overlooked, aspect of EQA is the agent's meta-cognitive calibration—its ability to assess its own uncertainty. The AbstainEQA benchmark evaluates an agent's capacity to recognize when it possesses insufficient information due to occlusions or underspecified instructions⁶⁷. Human operators instinctively pause and ask for clarification when instructions are ambiguous; however, standard VLA models often commit prematurely, confidently executing incorrect actions or providing hallucinated answers⁶⁶. Extensive evaluations reveal that even advanced VLMs achieve an abstention recall of only 42.79%, far below the human baseline of 91.17%, underscoring a significant deficiency in uncertainty-aware autonomous exploration⁶⁷.

Challenges and Prospects

While the integration of foundation models has dramatically advanced embodied intelligence, the transition from simulated benchmarks to ubiquitous real-world deployment faces several severe engineering and theoretical obstacles.

Data Scarcity and the Sim-to-Real Gap

Unlike natural language or static images, which can be scraped from the internet in the trillions of tokens, high-quality robotic trajectory data is exceedingly expensive and time-consuming to curate². Collecting kinesthetic teaching data requires human operators to manually guide robotic arms through precise movements, creating a severe scaling bottleneck²⁵.

Consequently, there is an aggressive push toward leveraging synthetic data generated within high-fidelity physics simulators. However, policies trained exclusively in simulation frequently

fail in the real world due to the "sim-to-real gap"—minor discrepancies in friction, lighting, sensor noise, and object mass¹⁹. Frameworks generating coded, logic-bound simulations, alongside large-scale video prediction world models, are actively attempting to bridge this gap by exposing training algorithms to highly randomized, physically accurate hypothetical scenarios during post-training phases⁴⁷.

Systems and Edge Deployment Constraints

Deploying massive neural architectures on physical robots is fundamentally a systems engineering challenge, bounded by strict size, weight, and power (SWaP) constraints⁶⁹. High-frequency, closed-loop autonomy requires perception and action matrices to be calculated with near-zero latency⁷¹. A monolithic 7-billion-parameter VLA executing an autoregressive policy can easily saturate the memory bandwidth (LPDDR channels) of embedded edge devices, resulting in severe thermal throttling and catastrophic frame drops⁷¹. The dense attention computation required by transformers directly conflicts with the localized SRAM capacities of mobile platforms⁷¹. To achieve viability, future architectures must lean heavily into hardware-aware quantization, token pruning, and linear-complexity sequence models. The RoboMamba architecture demonstrates this potential by integrating vision encoders with Mamba State Space Models (SSMs), achieving robust reasoning and inference speeds three times faster than transformer-based VLAs without sacrificing task accuracy³².

Knowledge Retention and Representation Degradation

A pernicious side effect of fine-tuning pre-trained VLMs on domain-specific robotic action data is catastrophic forgetting, also termed representation degradation⁷³. When models are optimized heavily for low-level motor output, the delicate alignment between vision and language—learned during massive internet-scale pre-training—begins to degrade. This loss of generalized semantic capability prevents the robot from adapting to novel visual conditions. To combat this, techniques such as Module-Wise Proximity Scheduling (MAPS) have been developed⁷⁴. MAPS dynamically restricts the gradient updates applied to the visual encoder during action fine-tuning, forcing the model to retain its pre-trained visual priors while allowing the later language and action heads to adapt freely. This selective relaxation preserves the broad visual comprehension of the base model while successfully acquiring task-specific kinematic skills⁷⁴.

Safety, Reliability, and Hallucinated Physics

As autonomous systems increasingly rely on internal world models to generate hypothetical trajectories, a unique threat surface emerges²⁰. Because world models are generative, minor initial errors compound exponentially across multi-step rollouts, creating hallucinations that diverge completely from physical reality¹⁹. Furthermore, these systems are highly susceptible to adversarial manipulation and reward hacking. If an adversarial patch is applied to a physical object, it can subtly shift the model's latent representation, causing the world model to confidently hallucinate an empty workspace or an open door where an obstacle exists²⁰. The action policy then executes a catastrophic collision based on this false simulation²⁰.

Additionally, if a learned reward head in a world model contains a local optimum, the agent may discover a non-productive motion sequence that tricks the internal simulator into outputting high success scores, trapping the robot in an endless loop of hallucinated success²⁰. Standard software security frameworks lack mechanisms to verify the physical truth of generated latent states, highlighting an urgent need for physics-informed verification protocols in embodied AI.

Conclusion

The evolution of Vision-Language-Action models represents a foundational leap toward artificial general intelligence, redefining the mechanisms by which machines interact with the physical world. By subsuming perception, semantic comprehension, and kinematic control within unified network architectures, VLAs successfully bypass the brittleness inherent in traditional, modular robotics. The rapid progression from basic autoregressive tokenization to high-frequency flow-matching policies demonstrates the field's commitment to achieving fluid, human-like dexterity. Concurrently, the integration of 3D spatial grounding, predictive world modeling, and active sensory exploration directly addresses the extreme complexities of dynamic, unstructured environments.

However, realizing the full potential of embodied foundation models requires surmounting significant obstacles. The dependency on scarce real-world trajectory data, the severe computational constraints of edge hardware, and the critical vulnerabilities surrounding representation degradation and generative hallucinations demand intensive, cross-disciplinary research. Moving forward, the synthesis of rigorous hierarchical logic with adaptive, unified sequence modeling will be paramount in creating autonomous systems that are not only profoundly capable but also inherently safe and reliable for ubiquitous physical deployment.

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